

Equilibrium Structural Work

Search, Spatial, Industry/Market, and Policy Counterfactuals
Empirical Methods — Week 8

Teaching deck draft

Week 8 question

- When do individual decisions interact through markets, prices, firms, or locations?
- What does equilibrium structure buy that partial equilibrium cannot?
- Which moments discipline the equilibrium primitives?
- How should policy counterfactuals and welfare be interpreted once markets adjust?

Equilibrium structure is worth the cost when incidence, allocation, or welfare depends on endogenous market response.

When do we need equilibrium structure?

Partial-equilibrium risk	Equilibrium object	Applied examples
Other agents' environments change	wages, prices, rents, vacancies	minimum wages, UI, housing policy
Matching changes	tightness, sorting, queues	job search, firm recruiting, platforms
Locations adjust	migration, commuting, housing costs	place policy, transit, local shocks
Firms respond strategically	markups, entry, pass-through	mergers, taxes, technology shocks
Flows reallocate	bilateral frictions and shares	trade, migration, commuting, applications

$$\frac{\partial E_i}{\partial a_j} \neq 0 \quad \text{for many } i \neq j$$

Search and matching equilibrium

$$\theta = \frac{v}{u}, \quad f(\theta) = \frac{m(u, v)}{u}, \quad q(\theta) = \frac{m(u, v)}{v}$$

$$\kappa = q(\theta)J$$

$$W - U = \beta S, \quad J = (1 - \beta)S$$

- Observed objects: unemployment, vacancies, transitions, separations, wages.
- Latent objects: matching technology, vacancy cost, surplus, bargaining or wage posting.
- Counterfactual gain: unemployment, vacancy, and wage incidence after firms respond.
- Main risk: matching and wage-setting assumptions carry the result.

$$V_{ij} = w_j - r_j - \tau_{ij} + A_j + \varepsilon_{ij}, \quad s_{ij} = \frac{\exp(V_{ij}/\sigma)}{\sum_{\ell} \exp(V_{i\ell}/\sigma)}$$

$$N_j(w, r, A, \tau) = H_j(r_j; \eta_j), \quad L_j(w, r, A, \tau) = D_j(w_j, Z_j)$$

$$dV_j = dw_j - dr_j + dA_j - d\tau_j$$

- Wages are not welfare when rents, amenities, and commuting costs adjust.
- Housing supply and mobility frictions govern incidence.
- Fit should include wages, rents, population or employment, and flows.

Industry and market equilibrium

$$q_j = D_j(p, x, \xi; \theta_d)$$

$$p_j - mc_j = - \left(\frac{\partial D_j(p, x, \xi; \theta_d)}{\partial p_j} \right)^{-1} D_j(p, x, \xi; \theta_d)$$

$$\frac{p_j - mc_j}{p_j} = -\frac{1}{\varepsilon_{jj}}, \quad (p^*, q^*, L^*) = \Psi(\theta_d, \theta_c, \Omega, Z)$$

- Demand and conduct imply prices, markups, output, and labor incidence.
- Useful for taxes, mergers, platform pricing, market power, and pass-through.
- Validation requires credible substitution patterns and price-cost implications.

Policy counterfactuals and welfare

$$x^*(\tau) = \Phi(x^*(\tau), \tau; \theta)$$

$$W(\tau) = \sum_i \omega_i U_i(x^*(\tau), \tau; \theta) + \sum_f \Pi_f(x^*(\tau), \tau; \theta) + R(x^*(\tau), \tau)$$

$$\Delta W = W(\tau_1) - W(\tau_0)$$

- The fixed point collects wages, vacancies, rents, prices, shares, entry, or flows.
- Welfare depends on distributional weights, profits, rents, and fiscal terms.
- Report counterfactual distance from support and sensitivity to closure assumptions.

Computational and identification burdens

Approach	Intuition	Risk
Nested fixed point	solve equilibrium inside parameter search	slow and opaque
MPEC	estimate with equilibrium constraints directly	numerical delicacy
Calibration-estimation hybrid	estimate some objects, calibrate others	headline result may rest on calibration
Simulation	approximate heterogeneous equilibrium response	seed, tolerance, and approximation error

$$\hat{m}^{data} \iff m^{model}(\theta, x^*(\theta))$$

- Name which moments identify which primitives.
- Show targeted and untargeted fit.
- Report solver, tolerances, starts, convergence, and sensitivity.

Gravity-style equilibrium models

$$M_{ij} = A_i B_j \tau_{ij}^{-\sigma}$$

- The object is a bilateral flow: trade, migration, commuting, applications, or worker allocation.
- The structural object is a friction or resistance term that governs reallocation.
- Gravity belongs here when flows close an equilibrium with wages, prices, rents, or employment.
- It should remain subordinate to the broader question: how does equilibrium incidence work?

Diagnostic question: does the model only describe flows, or does it solve the market response that matters for welfare?

Reproduce

- Hsieh-Moretti-style spatial equilibrium
- wages, rents, population
- housing relaxation counterfactual
- output and welfare table

Diagnose

- housing elasticity
- mobility elasticity
- targeted vs closure objects
- validation memo

Transfer

- BLP/Nevo-style market exercise
- demand elasticity
- markup recovery
- cost-shock pass-through

Takeaways

- Equilibrium structure is a research choice, not a badge of sophistication.
- Search, spatial, and market models differ in economics but share an empirical architecture.
- The model must say which equilibrium margin matters and which moments discipline it.
- Counterfactuals are credible only when fit, support, computation, and closure are transparent.
- The best equilibrium papers make the gains and burdens visible in the same language.